

Rename link to time_zone_link

[Addresses DE 346](#)

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Introduction

The LEWG in Belfast requested that I write a paper which provides wording for [DE 346](#) as modified by the discussion in LEWG. The LEWG chose the name `time_zone_link` as a replacement for `link`, and specifically chose not to change the associated data member of `tzdb` links.

Wording

1. Modify the synopsis in `[time.syn]` as indicated:

```
...
// 27.11.9, class time zone link
class time zone link;

bool operator==(const time zone link& x, const time zone link& y);
strong_ordering operator<==(const time zone link& x, const time zone link& y);
...
```

2. Modify the synopsis in `[time.zone.db.tzdb]` as indicated:

```
namespace std::chrono {
    struct tzdb {
        string          version;
        vector<time_zone> zones;
        vector<time zone link> links;
        vector<leap>    leaps;

        const time_zone* locate_zone(string_view tz_name) const;
        const time_zone* current_zone() const;
    };
}
```

3. Modify `[time.zone.db.tzdb]/p2` as indicated:

```
const time_zone* locate_zone(string_view tz_name) const;
```

~~2 Returns: If a `time_zone` is found for which `name() == tz_name`, returns a pointer to that `time_zone`. Otherwise if a `link` is found for which `tz_name == link.name()`, then a pointer is returned to the `time_zone` for which `zone.name() == link.target()`.~~

- If a time zone, tz, is found for which tz.name() == tz_name, a pointer to tz;
- otherwise, if a time zone link, tz l, is found for which tz l.name() == tz_name, then a pointer to the time zone, tz, for which tz.name() == tz l.target().

[Note: A time zone link specifies an alternative name for a time_zone. — end note]

4. Modify the title of [time.zone.link] as indicated:

Class time zone link

5. Modify [time.zone.link.overview] as indicated:

```
namespace std::chrono {
    class time zone link {
    public:
        time zone link(time zone link&&) = default;
        time zone link& operator=(time zone link&&) = default;

        // unspecified additional constructors

        string_view name() const noexcept;
        string_view target() const noexcept;
    };
}
```

1 A time zone link specifies an alternative name for a time_zone. time zone links are constructed when the time zone database is initialized.

6. Modify [time.zone.link.members]/p2 as indicated:

```
string_view target() const noexcept;
```

2 *Returns:* The name of the time_zone for which this time zone link provides an alternative name.

7. Modify [time.zone.link.nonmembers] as indicated:

```
bool operator==(const time zone link& x, const time zone link& y) noexcept;
```

1 *Returns:* x.name() == y.name().

```
strong_ordering operator<=>(const time zone link& x, const time zone link& y) noexcept;
```

2 *Returns:* x.name() <=> y.name().